

$$\begin{aligned}
& \frac{1}{6n^2} \left(\frac{1}{81} \sum_p g_1^4 LF_{3,0}[m_d^P] \delta_{i1i2} - \frac{5}{216} \sum_p g_1^4 LF_{4,-1}[m_d^P] \delta_{i1i2} + \frac{4}{405} \sum_p g_1^4 LF_{5,-2}[m_d^P] \delta_{i1i2} + \right. \\
& \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_e^P] \delta_{i1i2} - \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_e^P] \delta_{i1i2} + \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_e^P] \delta_{i1i2} + \\
& \frac{1}{54} \sum_p g_1^4 LF_{3,0}[m_l^P] \delta_{i1i2} - \frac{5}{144} \sum_p g_1^4 LF_{4,-1}[m_l^P] \delta_{i1i2} + \frac{2}{135} \sum_p g_1^4 LF_{5,-2}[m_l^P] \delta_{i1i2} + \\
& \frac{1}{162} \sum_p g_1^4 LF_{3,0}[m_q^P] \delta_{i1i2} - \frac{5}{432} \sum_p g_1^4 LF_{4,-1}[m_q^P] \delta_{i1i2} + \frac{4}{405} \sum_p g_1^4 LF_{5,-2}[m_q^P] \delta_{i1i2} + \\
& \frac{4}{81} \sum_p g_1^4 LF_{3,0}[m_u^P] \delta_{i1i2} - \frac{5}{54} \sum_p g_1^4 LF_{4,-1}[m_u^P] \delta_{i1i2} + \frac{16}{405} \sum_p g_1^4 LF_{5,-2}[m_u^P] \delta_{i1i2} + \\
& \frac{1}{54} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} + \frac{1}{36} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} - \frac{4}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{18} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,1,0}[m_1, m_d^P] - \frac{1}{9} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{3,1,-1}[m_1, m_d^P] + \\
& \frac{1}{18} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{4,1,-2}[m_1, m_d^P] + \frac{1}{2592} g_1^4 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} + \frac{1}{2592} g_1^4 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i1}] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{96} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i1}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{96} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i2}] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{96} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i2}] \delta_{i1i2} + \frac{2}{3} g_3^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,1,0}[m_3, m_d^P] - \\
& \frac{4}{3} g_3^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{3,1,-1}[m_3, m_d^P] + \frac{2}{3} g_3^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{4,1,-2}[m_3, m_d^P] + \\
& \frac{1}{54} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i1}] \delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i1}] \delta_{i1i2} - \\
& \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i1}] \delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{54} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i2}] \delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i2}] \delta_{i1i2} - \\
& \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i2}] \delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{18} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,1,0}[m_d^P, \tilde{\mu}] - \frac{1}{36} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{2,2,-1}[m_d^P, \tilde{\mu}] - \\
& \frac{1}{36} g_1^2 \overline{y}_d^{i2p} y_d^{i1p} LF_{3,1,-1}[m_d^P, \tilde{\mu}] + \frac{1}{18} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{2,2,0}[m_d^r, m_q^P] \delta_{i1i2} - \\
& \frac{1}{36} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,2,-1}[m_d^r, m_q^P] \delta_{i1i2} + \frac{1}{18} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{2,2,0}[m_e^r, m_l^P] \delta_{i1i2} - \\
& \frac{1}{36} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,2,-1}[m_e^r, m_l^P] \delta_{i1i2} - \frac{1}{18} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,1,0}[m_l^P, m_e^r] \delta_{i1i2} - \\
& \frac{1}{18} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,2,-1}[m_l^P, m_e^r] \delta_{i1i2} + \frac{1}{8} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{4,1,-1}[m_l^P, m_e^r] \delta_{i1i2} - \\
& \frac{1}{18} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{5,1,-2}[m_l^P, m_e^r] \delta_{i1i2} - \frac{1}{18} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,1,0}[m_q^P, m_d^r] \delta_{i1i2} - \\
& \frac{1}{18} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,2,-1}[m_q^P, m_d^r] \delta_{i1i2} + \frac{5}{24} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{4,1,-1}[m_q^P, m_d^r] \delta_{i1i2} - \\
& \frac{1}{6} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{5,1,-2}[m_q^P, m_d^r] \delta_{i1i2} - \frac{1}{36} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{2,2,0}[m_q^P, m_u^r] \delta_{i1i2} + \\
& \frac{1}{72} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,2,-1}[m_q^P, m_u^r] \delta_{i1i2} - \frac{1}{1296} g_1^4 LF_{2,1,0}[m_q^{i1}, m_1] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{3,1,-1}[m_q^{i1}, m_1] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i1}, m_2] \delta_{i1i2} + \\
& \frac{1}{96} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i1}, m_2] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i1}, m_3] \delta_{i1i2} + \\
& \frac{1}{54} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i1}, m_3] \delta_{i1i2} - \frac{1}{1296} g_1^4 LF_{2,1,0}[m_q^{i2}, m_1] \delta_{i1i2} + \\
& \frac{1}{2592} g_1^4 LF_{3,1,-1}[m_q^{i2}, m_1] \delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i2}, m_2] \delta_{i1i2} + \\
& \frac{1}{96} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i2}, m_2] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i2}, m_3] \delta_{i1i2} + \\
& \frac{1}{54} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i2}, m_3] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{y}_u^{i2p} y_u^{i1p} LF_{2,1,0}[m_u^P, \tilde{\mu}] + \\
& \frac{1}{18} g_1^2 \overline{y}_u^{i2p} y_u^{i1p} LF_{2,2,-1}[m_u^P, \tilde{\mu}] + \frac{1}{18} g_1^2 \overline{y}_u^{i2p} y_u^{i1p} LF_{3,1,-1}[m_u^P, \tilde{\mu}] + \\
& \frac{1}{36} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,1,0}[m_u^r, m_q^P] \delta_{i1i2} + \frac{1}{36} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,2,-1}[m_u^r, m_q^P] \delta_{i1i2} + \\
& \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{4,1,-1}[m_u^r, m_q^P] \delta_{i1i2} - \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{5,1,-2}[m_u^r, m_q^P] \delta_{i1i2} - \\
& \frac{5}{72} g_1^4 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{1}{18} g_1^4 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2}$$